

Characterisation of dissimilar joints in laser welding of steel–kovar, copper–steel and copper–aluminium

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Abstract

Joining of dissimilar materials is one of the challenging tasks facing modern manufacturers. Dissimilar joining technologies find applications in many sectors including microelectronics, medical, optoelectronics and microsystems. The tiny geometry of the joints and the different optical and thermal properties of the materials makes laser welding one of the most suitable production methods. This paper presents the results of an investigation into laser welding of dissimilar metals without filler materials using a 350 W pulsed Nd:YAG laser. The mixing behaviour of the materials in the fusion zone, the microstructure, the presence of defects, hardness and residual stress of the joints were all investigated. © 2004 Elsevier B.V. All rights reserved.

Keywords: Laser welding; Dissimilar materials; Microstructure; Hardness; Residual stress; X-ray inspection

1. Introduction

Joints between dissimilar metals are commonly used in the power generation, chemical, petrochemical, nuclear and electronics industries for the purpose of tailoring component properties or weight reduction. More recently, laser-welding technologies have been successfully used to manufacture hybrid microsystems consisting of different materials [1–3].

The weldability of dissimilar metals is determined by their atomic diameter, crystal structure and compositional solubility in the liquid and solid states. Diffusion in the weld pool often results in the formation of intermetallic phases, the majority of which are hard and brittle and are thus detrimental to the mechanical strength and ductility of the joint. As no filler materials are used, the formation of intermetallic compounds is dependent on the interaction of the joining materials and the welding parameters. In laser welding, the risk of segregation and excessive formation of brittle phases is significantly reduced due to the high cooling and solidification rates. However, high temperature gradients may result in martensitic reactions leading to excessive hardness in the fusion zone [4,5]. In addition, the high thermal gradients

also promote high residual stresses, which in turn affect the integrity of the joints.

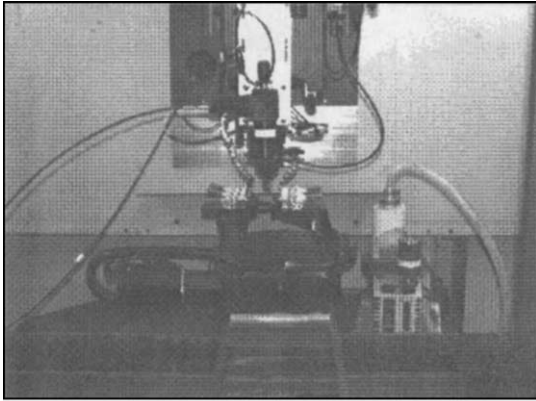
Aluminium and copper are difficult to melt with lasers due to their high reflectivity and high thermal conductivity. In laser welding of butt joints, precise control of the laser focus position with respect to the joint interface is essential to achieve a sound weld. Depending on the reflectivity and the melting point of the joining materials, an optimal melting ratio is typically achieved by off-setting the beam focus from the joint interface. The focus position varies with the laser operating mode (cw or pulsed), the laser wavelength (CO₂ or Nd:YAG laser) and the output power of the laser used. Positive results have been reported when the laser beam is positioned completely or mainly within the lower reflective joining metal. In this case, a limited amount of the highly reflective part is melted by the highly absorptive molten part via heat transfer [6,7].

2. Experimental

A pulsed Nd:YAG laser system (Lumonics JK 702H) with a maximum average output power of 350 W was used in this study. The pulse width can be varied between 0.5 and 20 ms at a maximum pulse frequency of 500 Hz. The full beam divergence was 14 mrad. The pulsed Nd:YAG laser has adjustable pulse shapes which offers high flexibility in

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Laser welding head with a clamping device

Fig. 1. Laser welding head with a clamping device.

optimising the weld parameters to achieve defect free joints. The laser beam was delivered to the work piece by mirrors and focused by 76 mm lens. The laser spot size on the work piece was approximately 0.75 mm. The experimental set up with a clamping device is shown in Fig. 1.

Laser welding experiments were performed on three different metal combinations (Kovar–tool steel, copper–tool steel and copper–aluminium alloy). Commercially available medium carbon tool steel, oxygen-free copper and aluminium 4047 were used. Kovar is a Fe–Ni–Co alloy with very low thermal expansion coefficient. The melting point of Kovar is about 1450 °C. Aluminium 4047 is an Al–Si alloy with low crack sensitivity due to its narrow solidification temperature range (565–575 °C).

The most important thermophysical properties of the corresponding metals are shown in Table 1. Although these values refer to pure metals and some properties are temperature dependent, the data provided a basic reference for assessing the weldability and deriving the welding strategy. Copper conducts the heat energy up to 10 times faster than steels. Copper and aluminium are also highly reflective compared to steels. In both copper and aluminium only a small amount of the laser energy is absorbed and this energy dissipates rapidly into the bulk material, making it difficult to maintain a molten pool. Successful welding of copper and aluminium requires a high pulse peak power and a short pulse duration.

Air gaps in the joints, as well as misalignment of the plates, can significantly affect the weld quality. In order to obtain a nearly zero gap between the two joining partners (i-butt joints), the 1 mm thick samples, obtained in sheet form were cut with a high machining tolerance into blocks of 75 mm × 20 mm × 1 mm by EDM. The samples were

cleaned and aligned with the aid of a specially designed clamping device.

The melting ratio and the weld quality depend on both the material and the welding parameters such as spot size, focus position, welding speed, pulse duration, pulse peak power and welding speed. In this study, laser welding was carried out in conduction mode at power densities below 10⁷ W/cm² to avoid keyhole formation.

Two etchants were developed to reveal the microstructure of the welds. The first etchant composition was 50 ml HCl + 10 ml HNO₃ + 5 g FeCl₃ + 100 ml H₂O. A composition of 40 ml HCl + 5 g CuCl₂ + 30 ml H₂O + 25 ml alcohol was used as the second etchant. Etching was done by 3–10 swabbing strokes. The hardness of the welds was measured using a Mitutoyo micro-hardness tester with the following parameters: 10 g load, 10 s duration of impact and 3 μm/s loading speed. A Rigaku Gigerflex X-ray microstress analyser and a Fein Focus FIP 400 X-ray imaging machine were used to investigate the residual stress and the internal features of the joints, respectively. It should be noted that determination of the exact value of residual stress within this region was difficult due to a number of factors such as surface roughness, no-known materials constants and changing composition all of which will effect the measured residual stress. The residual stress was determined using Cr Kα radiation and the Ψ^2 method with the following parameters $2\theta_{\text{Cu}} = 127^\circ$, $2\theta_{\text{Fe}} = 156^\circ$, $2\theta_{\text{Ni}} = 127^\circ$, $\Psi = -20, -15, -10, -5, 0, 5, 10, 15$ and 20°).

EDX element analysis was performed with a Jeol 6340F field emission SEM equipped with an Oxford instruments ISI EDX system. Point counting and line-scan intensity profiles for Fe(Kα), Cu(Kα), Al(Kα), Co(Kα), Ni(Kα) and Si(Kα) across the weld at 0.2 mm from the top surface were carried out to determine the dilution rate and the size of the weld pool. In the current study, the laser beam was positioned on the steel plate in both the steel–Cu and steel–Al joints. Any melting of the copper and aluminium was through contact heat transfer rather than direct heating from the laser beam. However, the beam was positioned at the joint interface in the steel–Kovar welding. Melting of both materials was achieved directly from the laser irradiation.

3. Results and discussion

3.1. Kovar and tool steel

Laser welding of steel and Kovar resulted in significant intermixing of both materials within the fusion zone (Table 2

Table 1
Thermophysical properties of some pure materials at room temperature [8,9]

Metal	Melting point (°C)	Boiling point (°C)	Density (g/cm ³)	Conductivity (W/m °C)	Heat capacity (J/(g °C))	Absorptivity (%)
Fe	1536	2860	7.87	0.78	0.456	40
Al	660	2520	2.70	2.38	0.917	9
Cu	1083	2560	8.93	3.97	0.386	3

Table 2
Quantitative EDX element analysis

Material	Element (at.%)				
	Fe	Ni	Co	Mn	Si
Tool steel	96.9	–	–	1.6	1.5
Kovar	53.6	28.5	16.2	0.3	1.4
Fusion zone	76.4	13.5	7.8	0.7	1.6

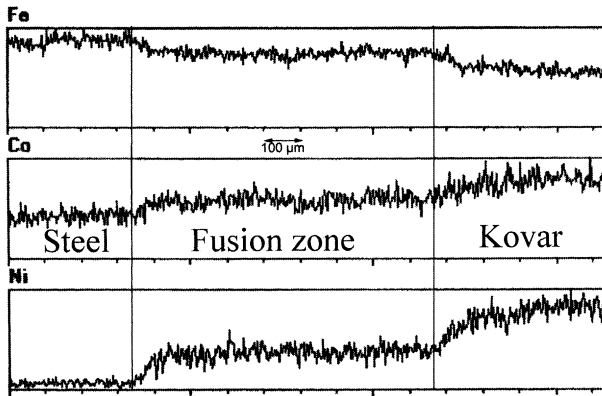


Fig. 2. Quantitative elemental distribution.

and Fig. 2). The weld pool dilution rate was approximately 50%. The EDX line-scan results showed that homogeneous mixing occurred. EDX analysis showed dissolution of Ni, Co and Mn into the weld pool and almost no composition variation indicating a complete mixing.

No hot cracking of the joint was observed, possibly due to the positive influence of Mn and Si dissolution into the weld. Cross-sections of weld seam exhibited strong convection currents, which were formed during the melting phase and were frozen due to the rapid solidification (Fig. 3). Large pores were trapped at the bottom of the weld at short laser pulses (<7 ms) or at high welding speeds (>500 mm/min).

The porosity of the joint was investigated using X-ray imaging. Fig. 4a shows a clear weld line at the interface of two materials. The bright line at the interface was due to

a gap at the bottom of the two metal plates. Between the two metals, the weld pool could be clearly distinguished in the X-ray image. The lighter contrast within the weld pool compared to the base metals indicates that the material was thinner than the base steel sheet. Close inspection of the contrast variation shows a uniform thickness with very little porosity.

The weld in Fig. 4b shows a large amount of elliptical shaped pores. The extensive porosity within the weld pool makes determination of the weld thickness profile difficult. The pores followed the shape of the weld front formed during the pulsed laser welding process. The width of the weld pool increased from left to right due to the transient development of the temperature at the beginning of the weld seam.

Examination of the porosity showed clear trends associated with welding speed. Increasing the welding speed appeared to decrease the number of pores, while increasing their average size (Figs. 5 and 6). Increasing the welding speed resulted in a decrease in the melting depth. As lesser time, on an average, is needed for a pore to move towards the surface, if the molten pool is shallow, the number of pores is reduced. The increase in pore size with welding speed is likely to be associated with the greater instability of the molten pool due to the higher peak power of the laser pulse used at faster welding speeds. Within the weld pool isolated islands with dark contrast were also observed. These regions are likely to be associated with a locally high atomic number or locally thick regions.

The morphology of Kovar microstructure shows fine eutectic distributed α -grains, whilst a dispersion of fine particles of alloy carbides in a matrix of ferrite was observed in tool steel. Fine grain structures of pearlite and martensite were observed in the weld pool as a result of high cooling rates.

A sharp increase in hardness was observed in the weld pool as compared to the base materials (Fig. 7). The hardness values on both sides of the graphs decreased sharply at a distance of less than 1 mm from the centre of the weld. It could also be seen that the core hardness of the tool steel is slightly higher than Kovar. One way to reduce the high

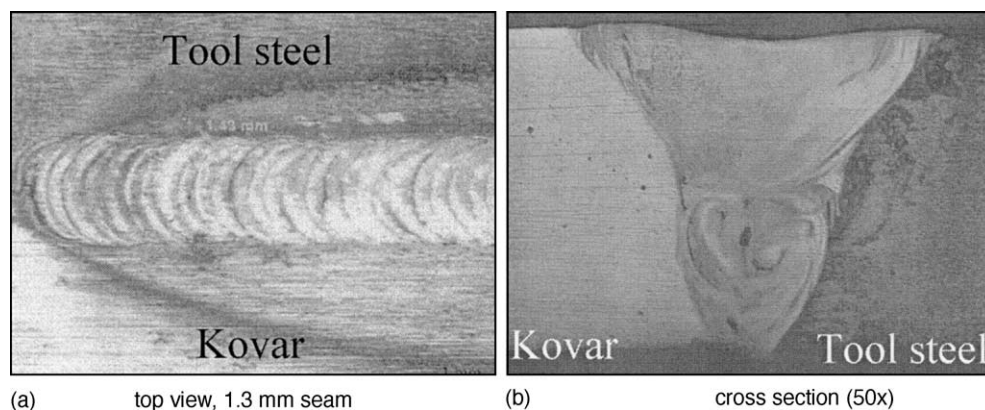


Fig. 3. Weld sample (30 Hz, 320 W, 5 ms and 150 mm/s).

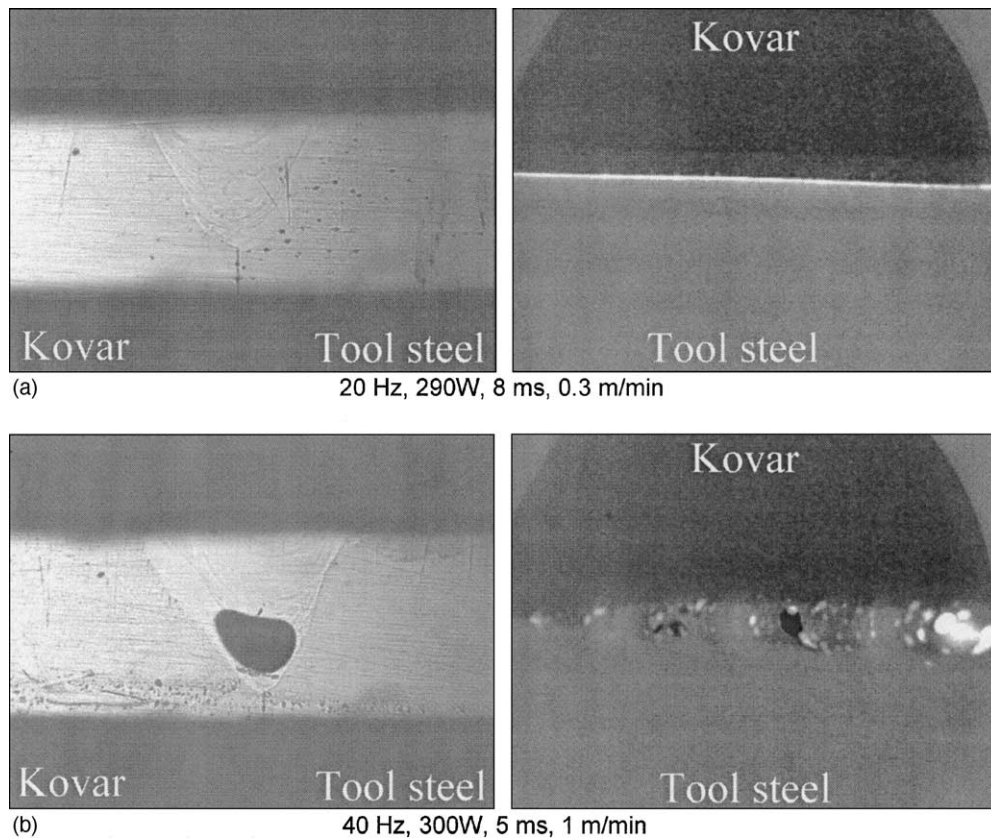


Fig. 4. Cross-sections and X-ray photographs.

hardness within the weld pool would be to preheat both plates before welding.

The residual stress increased within the steel plates as the welded area was approached, (Fig. 8). On entering the weld pool a small dip in the residual stress was observed. Within the weld pool the residual stress increased rapidly to highly tensile values. As the weld metal heats up it undergoes thermal expansion, however it is restrained by the surrounding base metal. On cooling, the weld metal contracts and is again

restrained by the base metal resulting in tensile stresses in the weld pool. Due to the restraining action of the base metal and its much larger dimensions a small compressive stress is induced away from the weld pool. Consequently, after cooling a residual tensile stress existed in the weld metal and in the adjacent base metal, while compressive stresses were present in the areas farther away from the weld. Within the fusion zone and the Kovar close to the weld, high magnitude residual stresses were observed.

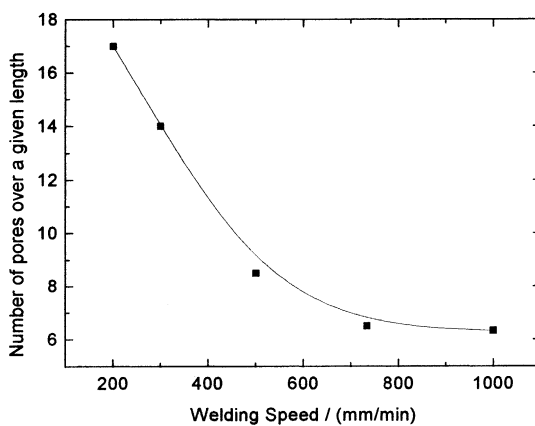


Fig. 5. Effect of welding speed on weld porosity and number of pores (300 Hz, 290 W and 5 ms).

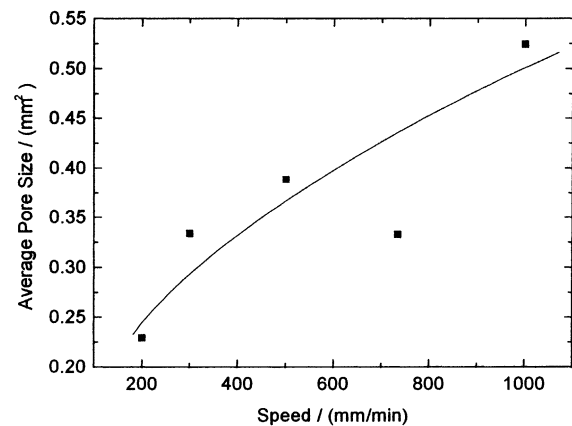


Fig. 6. Effect of welding speed on weld porosity and pore size (300 Hz, 290 W and 5 ms).

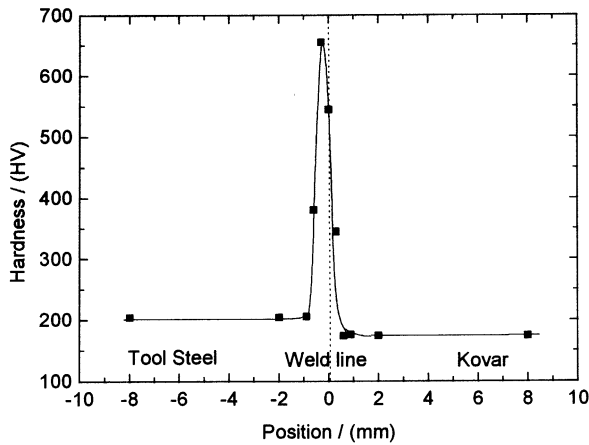


Fig. 7. Hardness distribution (9 ms, 20 Hz, 230 W and 0.2 m/min).

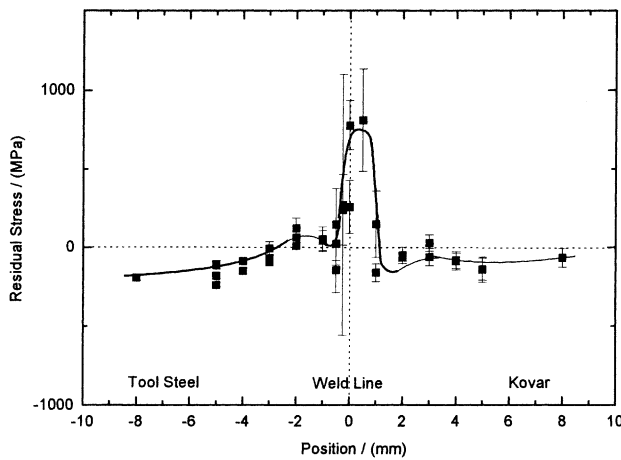


Fig. 8. Average residual stress variation of the welds.

The residual stress at the centre of the weld is close to that of the yield strength, (estimated from hardness measurements) micro of the weld metal. At the edge of the weld pool the residual stress rapidly reaches the normal value within the Kovar plate [10,11].

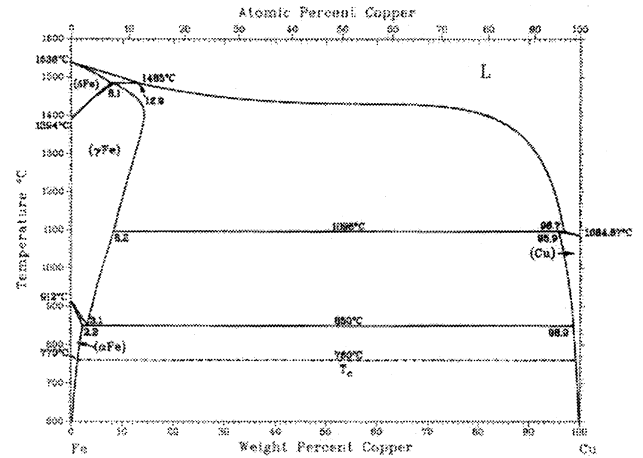


Fig. 9. Phase diagram Fe-Cu.

3.2. Copper and tool steel

In the field of power generation and transmission, cryogenics, electrical and electronics, copper-steel combinations are often used due to their high electrical conductivity and stiffness. The high thermal conductivity of copper tends to rapidly dissipate heat away from the weld leading to difficulties in reaching the melting temperature. The Fe-Cu phase diagram shows only very limited solubility of Cu in Fe, especially at low temperatures (Fig. 9). In addition, no low temperature Fe-Cu phase exists.

The major problem in welding copper to steel is hot cracking in the heat-affected zone of steel due to copper melting and penetrating into the grain boundaries of solid steel. The application of laser welding, particularly under pulse mode, is well suited to avoid this problem due to the rapid and intensive energy delivery.

In the present study, butt joints were created by focusing the laser beam 0.2 mm into the steel. Most of the laser energy was absorbed in the steel. A limited amount of copper was melted by the molten steel as its melting temperature is

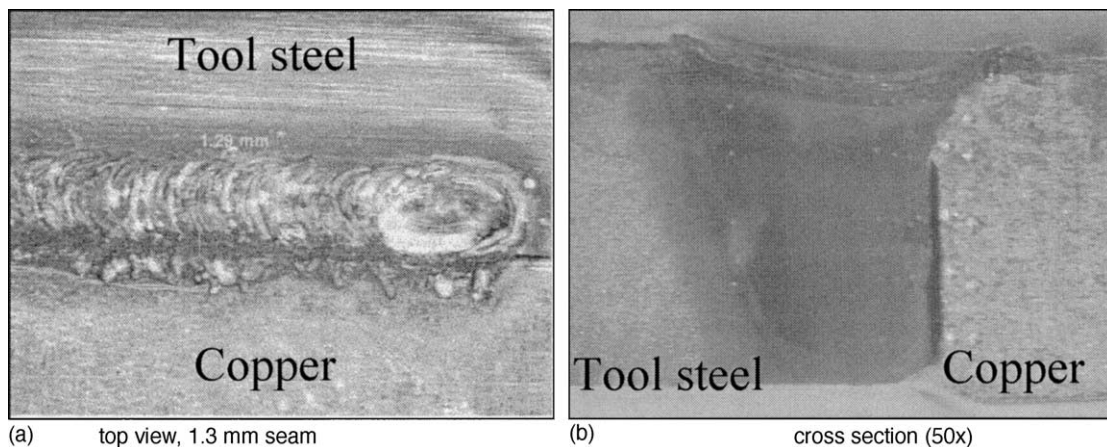


Fig. 10. Laser welded sample (30 Hz, 320 W, 5 ms, pulse shaping and 150 mm/min).

Table 3
Element analysis (relative values)

Material	Element (at.%)	
	Fe	Cu
Tool steel	100	—
Copper	—	100
Fusion zone	98.09	1.91

approximately 360 °C lower than that of steel. The amount of copper dissolved in the molten steel was very limited (<2%) and hence, no microcracks appeared at the weld (Table 3 and Fig. 10). It can be seen from Fig. 11, that the intermixing zone between steel and copper is less than 70 µm.

Although the steel plate was completely melted, a complete metallurgical bond could not be obtained at the interface between the Cu-plate and the weld metal. The main reason was associated with the high thermal conductivity of copper. The shorter molten time at the bottom compared to the top of the weld pool and the contamination of the joint surfaces may also have hindered proper weld penetration. Preheating or higher power density may allow improved joint integrity.

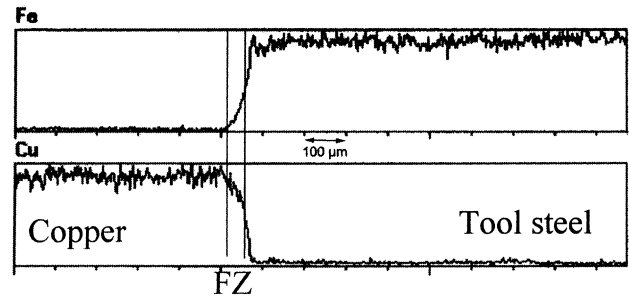
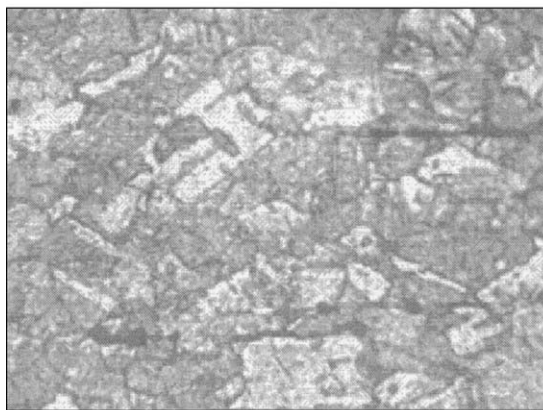


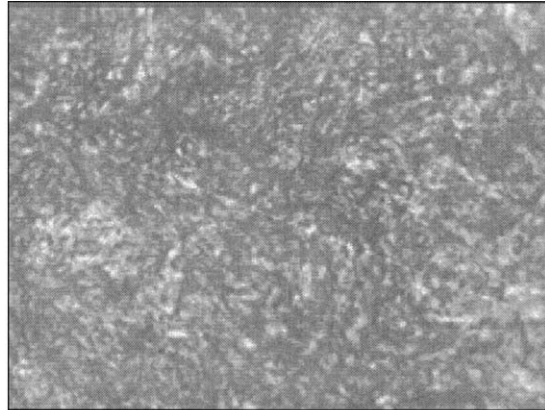
Fig. 11. Quantitative elemental distribution.

The various microstructures are shown in Fig. 12. Copper exhibited equiaxed grains and dispersion of Cu₂O particles. The microstructure of tool steel was made up of fine pearlite grains (see also Fig. 7). In the fusion zone, a dendritic microstructure with the presence of martensite was observed.

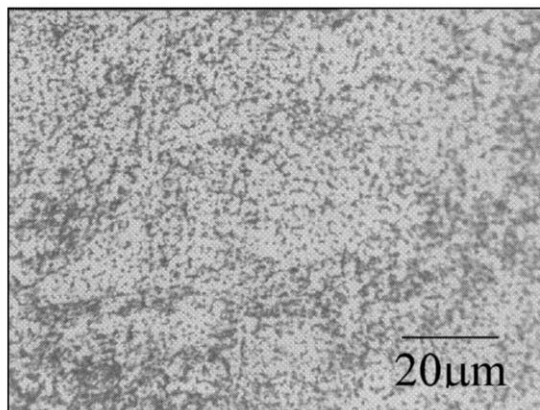
The hardness at different distance from the weld line is shown in Fig. 13. No significant change in hardness on steel side occurred. However, within the weld zone the hardness was over three times higher than that in the base metal.



(a) copper



(b) fusion zone



(c) tool steel

Fig. 12. Microstructure of copper/steel weld.

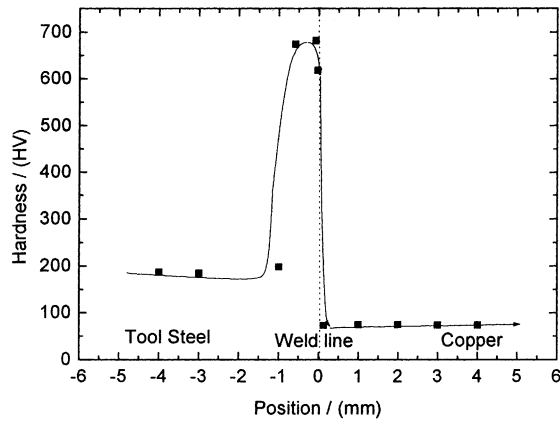


Fig. 13. Hardness as function of distance from joint interface.

X-ray imaging of the weld in Fig. 14a indicates extensive porosity. In addition, the very high contrast of the joint interface denoted a weak bond with copper plate. In the second weld (Fig. 14b), the porosity within the fusion zone was significantly reduced and the pores were relatively small. In addition the weld region appeared to have a much more uniform contrast suggesting that it is much smoother than the previous weld (Fig. 14a). The extensive porosity and the

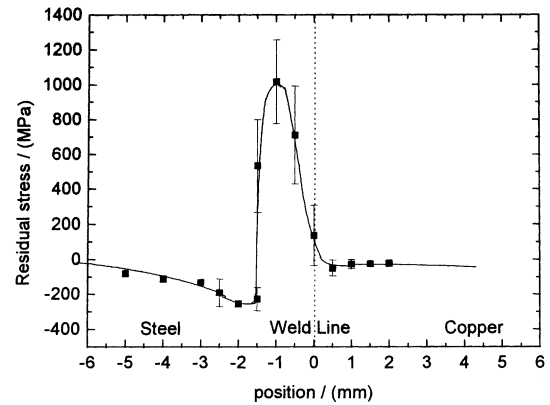


Fig. 15. Average residual stress variation of the welds.

high content at the joint interface were caused by a relatively large air gap between the two welded specimens.

The residual stress distribution across the welds was very similar between different samples (Fig. 15). The X-ray stress analysis technique was unable to resolve the stress variations generated by the different processing parameters used. In similar steel welds the residual stress at the centre of the weld pool has been reported to be close to the material yield

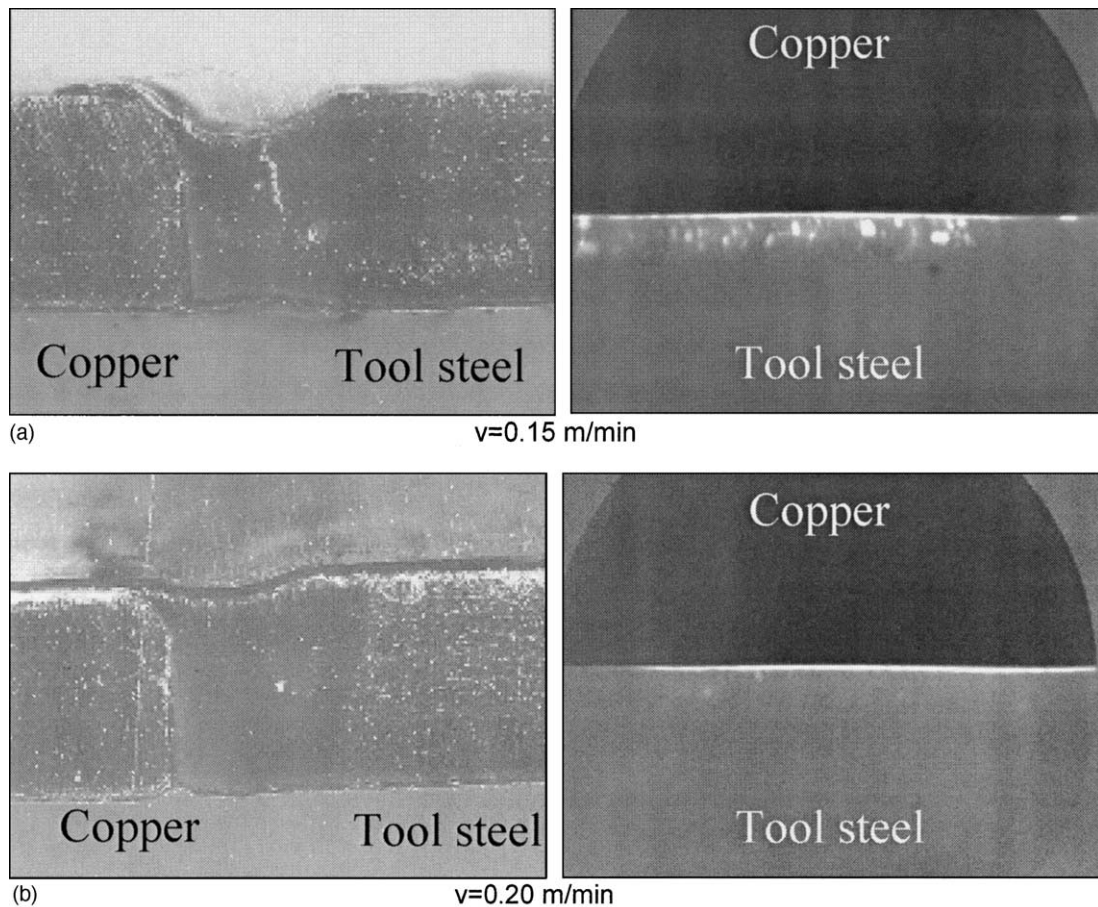


Fig. 14. Cross-section and X-ray photographs (325 W, 30 Hz, and 6 ms).

Table 4
Element analysis (relative values)

Material	Element (at.%)		
	Cu	Al	Si
Copper	100	—	—
Al 4047	—	70.98	29.02
Fusion zone	27.24	48.43	24.34

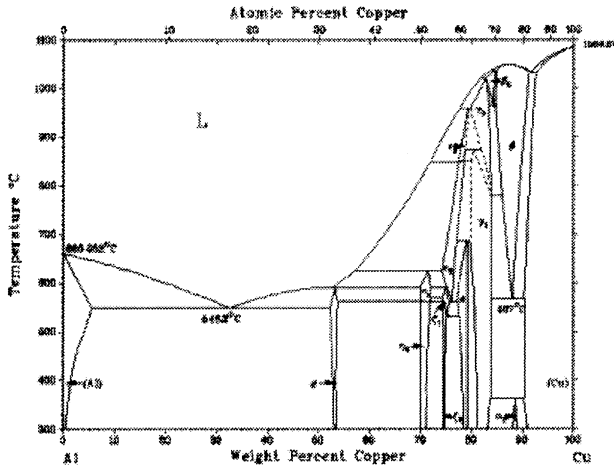


Fig. 16. Phase diagram Al–Cu.

strength [11]. A high tensile residual stress in this weld is therefore not surprising. It should be noted that tensile strength in the sheet material is about 800 MPa, which is lower than the measured residual stress. However, based on the hardness measurements, the tensile strength in the weld pool material may exceed 2000 MPa, which is well above the measured value of the residual stress.

3.3. Copper and aluminium 4047

Joints between aluminium and copper are often required in electrical components or solar collectors. From the phase

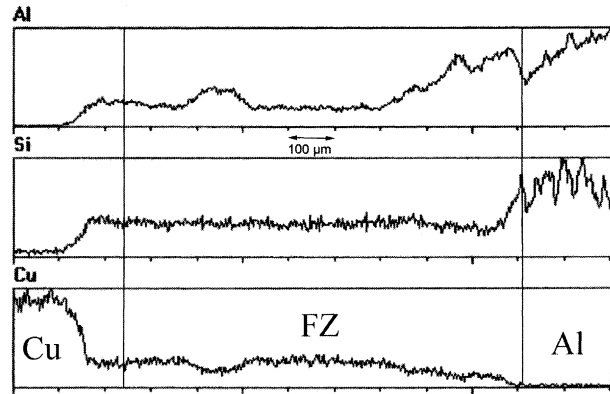


Fig. 18. Quantitative elemental distribution, line-scan of a weld cross-section.

diagram in Fig. 16, the maximum solid solubility of copper in aluminium is 5.65 wt.% Cu. At lower temperatures, the amount of copper that can remain in the aluminium solid solution, decreases drastically.

The weldability of copper to aluminium is regarded as relatively good. The Cu–Al phase diagram shows a wide range of Cu–Al phases that may be formed. In addition, non-equilibrium cooling conditions are known to promote the formation of a cascade of metastable phases. Under optimum processing parameters, crack-free welds of copper and Al 4047 were obtained. Attempts to weld copper with aluminium alloy 6061, a heat-treated Al–Mg–Si–Cu alloy with high mechanical strength, were not successful due to its poor weldability and high crack susceptibility. Solidification cracks were observed at the centre of copper–aluminium 4047 joints at welding speeds greater than 100 mm/min. However, sound welds with aluminium 4047 were obtained at lower welding speeds (Fig. 17).

Table 4 shows the fusion zone consisting of aluminium (approximately 48.4%), silicon (24.3%) and copper (27.2%). As the beam was focused 0.2 mm into aluminium (from the joint interface) and the absorptivity of copper was lower than aluminium alloy, melting took place more on the aluminium

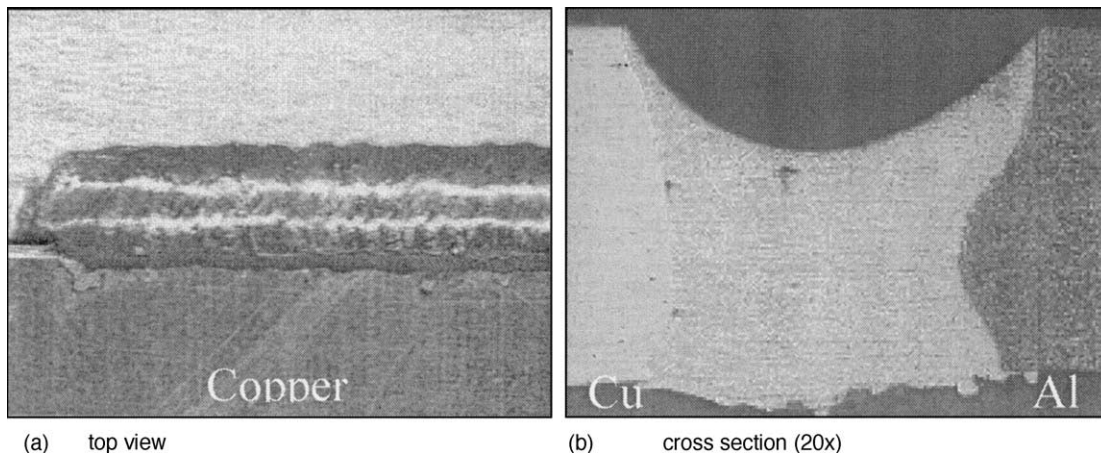


Fig. 17. Laser welded sample (1.3 mm seam, 15 Hz, 275 W, 8 ms, pulse shaping, and 75 mm/min).

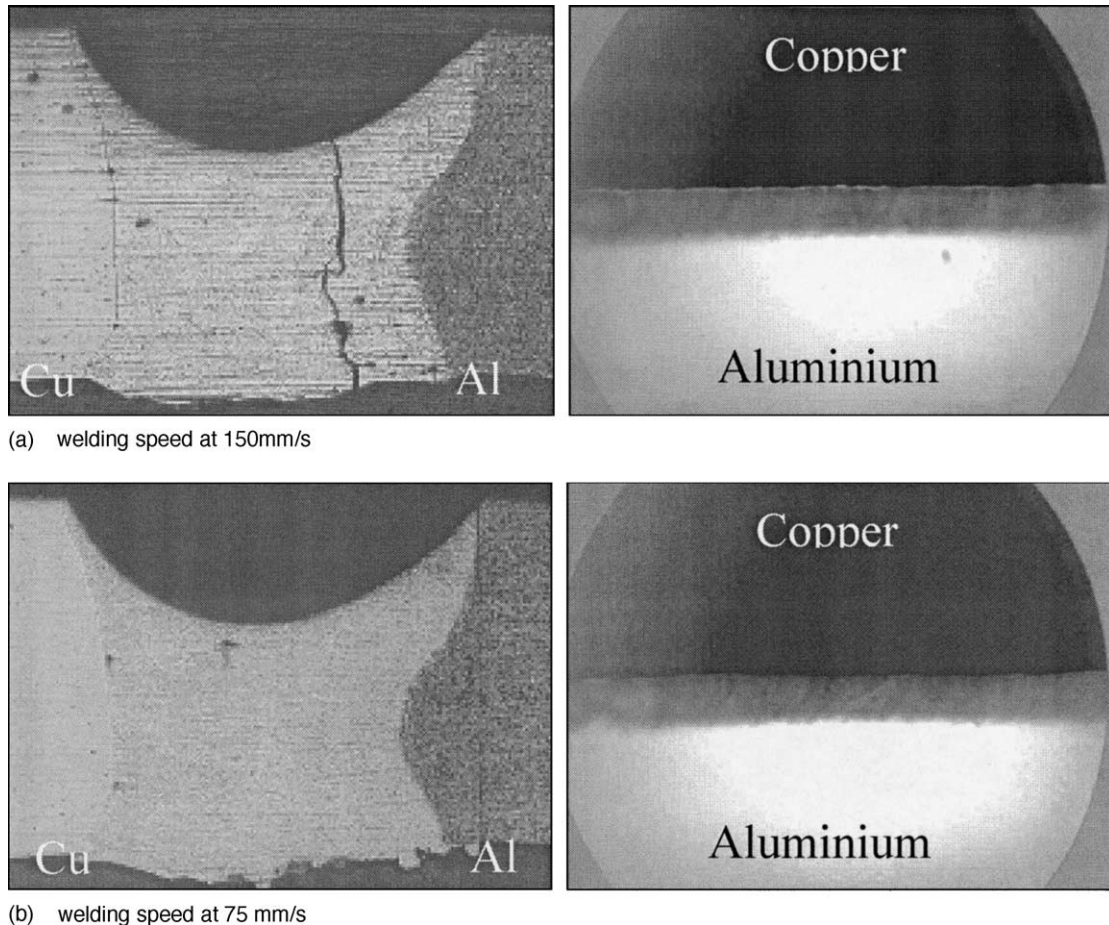


Fig. 19. Cross-section and X-ray photographs (15 Hz, 275 W, 8 ms and pulse shaping).

side. The aluminium fusion zone was diluted with approximately 27% of copper and 24% of silicon. The silicon was distributed homogeneously across the weld (Fig. 18). However, it can be seen that along the 1 mm weld seam, the composition of Al increased and the Cu decreased from the point which is 0.7 mm from the copper side. There was also

a fluctuation of these two elements at a distance of 0.35 mm from the copper side.

The X-ray image in Fig. 19a shows a relatively uniform weld with microcracks running at about 45° to the welding direction. Small isolated copper regions were deposited onto the aluminium plate. The X-ray image of the weld in Fig. 19b shows similar features. No major porosity could be detected within the bulk of the weld, (except at the beginning of the welding process).

Fig. 20 depicts the hardness distribution from the centre of the joint. The maximum hardness was measured at the centre of the weld. This high hardness may be associated with the presence of supersaturated solid solution. Alternatively, the hardness may be due to the formation of GP-zones or related structures.

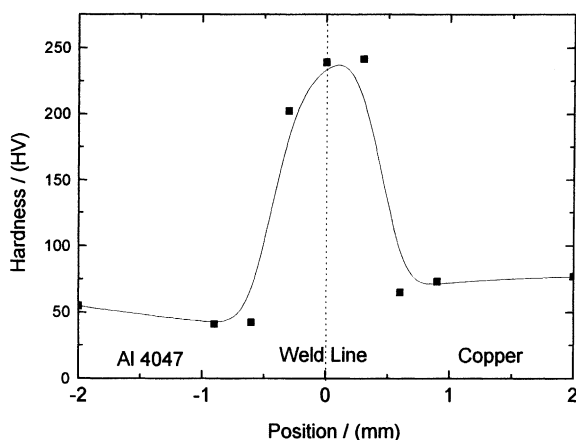


Fig. 20. Hardness distribution as function of distance.

4. Conclusions

Sound welds of three different materials combinations were produced by laser welding with a pulsed Nd:YAG laser. It has been shown, that controlling the melting ratio of metals is an important factor for defect-free welding of

dissimilar metals. Due to locally restricted energy input and high power density, laser welding permits a controlled heat distribution and a minimised interaction of the joining materials. Thus, the formation of brittle intermetallic phases could be avoided.

Besides tailoring the mechanical properties of a product, the results of the present study can be used for further investigations into joining of microsystem's components.

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